

Activity Guide — middle / module-05-dot-addition

IKS Curriculum

Table of Contents

Activity 01 — Column Drill (Multi-column)

Module: Bindu Paddhati · **Band:** Middle · **Day:** 4 · **Time:** 25 min

Type

Paired practice with self-correction

Goal

Each student solves 8 multi-column addition problems using the dot method, with the partner checking accuracy.

Materials per pair

- 1 printed problem sheet (8 problems, see below)
- 1 stopwatch
- 1 pen each

Set-up (teacher, before class)

- Print the problem sheet (you can vary the numbers — keep difficulty graded from 3×3-digit to 5×4-digit).

Instructions (student-facing handout)

Multi-column Drill — work in pairs.

1. Both partners solve all 8 problems individually (8 min). Don't look at each other's work.
2. Now swap sheets. Partner-A reads problem 1 aloud; partner-B re-solves it on a fresh sheet (no peeking at partner-A's answer). Compare answers. If they differ, work out which is correct.
3. Repeat for problems 2–8, alternating who reads.
4. Last 5 min: each student re-does any 3 problems they got wrong, silently.

Problem sheet

1.	234	2.	587	3.	429	4.	163
	356		342		718		274

+ 198	+ 925	+ 386	395 + 822
5. 2847 3576 4928 + 1235	6. 5683 2917 4256 + 1845	7. 38271 45982 36475 29384 + 41728	8. 58273 14916 62385 + 27548

[Answers (teacher copy):

1. 788 2. 1854 3. 1533 4. 1654
5. 12586 6. 14701 7. 191840 8. 163122]

Discussion prompts (last 2 min)

- Which problem took the longest? Why?
- Did your partner's solution differ from yours on any problem? Whose was right?
- Were the carries the most error-prone part?

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Pair is comparing answers, pointing at specific digits, debating	"I got 14702, you got 14701 — where's the difference?"
Needs intervention	One partner just copies the other's answers	Silence; "fine, whatever"

Safety / sensitivity

- No timed competition between pairs publicly. Pairs are self-pacing.

Differentiation

- **Tier up:** 5×5-digit problems (printed on a separate sheet).
- **Tier down:** 3×3-digit problems only; build up.

Clean-up

- Worksheets collected (for the teacher to spot-check before Day 5).

Activity 02 — Near-Base Mental Addition

Module: Bindu Paddhati · **Band:** Middle · **Day:** 7 · **Time:** 20 min

Type

Quick-fire mental math drill, individual + paired

Goal

Students add 2-digit numbers ending in 9, 19, 29, 49, 99 mentally using the “add the base, subtract the deficiency” pattern, in under 3 seconds per problem.

Materials per pair

- 1 printed drill sheet (20 problems below)
- 1 stopwatch

Instructions (student-facing handout)

Near-Base Mental Drill — work in pairs.

Round 1 (5 min, individual): Solve as many of the 20 problems as you can on the sheet. NO writing intermediate steps — write only the answer.

Round 2 (5 min, paired): Partner-A reads a problem aloud. Partner-B answers in under 3 seconds. Swap. Continue alternating through all 20.

Round 3 (5 min, mixed): Teacher displays 10 new problems on the board. Class shouts the answer in chorus.

Round 4 (5 min, quick-fire): Teacher calls a 2-digit number ending in 9 or 8, then any other 2-digit number; students whisper-answer to their neighbour.

Drill sheet (20 problems)

- | | |
|--------------|---------------|
| 1. $33 + 9$ | 11. $27 + 19$ |
| 2. $47 + 9$ | 12. $56 + 29$ |
| 3. $64 + 19$ | 13. $35 + 39$ |
| 4. $28 + 19$ | 14. $73 + 49$ |
| 5. $55 + 48$ | 15. $81 + 19$ |
| 6. $36 + 38$ | 16. $42 + 28$ |
| 7. $19 + 19$ | 17. $65 + 99$ |
| 8. $72 + 28$ | 18. $88 + 12$ |
| 9. $46 + 39$ | 19. $54 + 47$ |
| 10. $91 + 9$ | 20. $33 + 88$ |

Answer key (teacher copy)

- | | | | | | | | | |
|---------|---------|--------|--------|---------|---------|--------|---------|---------|
| 1. 42 | 2. 56 | 3. 83 | 4. 47 | 5. 103 | 6. 74 | 7. 38 | 8. 100 | 9. 85 |
| 10. 100 | 11. 46 | 12. 85 | 13. 74 | 14. 122 | 15. 100 | 16. 70 | 17. 164 | 18. 100 |
| 19. 101 | 20. 121 | | | | | | | |

Discussion prompts (after Round 4)

- Which type of problem was easiest? (Most students say +9 or +19 — single digit deficiency.)
- Which type took longer mental processing? (Often +48, +99 — bigger numbers requiring base-100 thinking.)
- Anyone discover a *new* shortcut while doing this?

Safety / sensitivity

- Same as Activity 01 — no public competition.

Differentiation

- **Tier up:** 3-digit near-base additions ($234 + 99$, $567 + 199$, etc.).
- **Tier down:** Stick to $+9$ and $+19$ problems for today. Build up.

Clean-up

- Worksheets kept by students for revision.

Project Brief

Project Brief — Module 5 Dot Addition (Middle)

Work sessions: Day 8 + Day 9 · **Presentation:** Day 10 · **Total marks:** 40 (project 30 + presentation 10)

The project

Each pair (or individual, with teacher approval) chooses ONE of three formats:

Format 1: 60-second teaching video

Make a short video that teaches the dot method to a Grade 4 student. Constraints: - Total length: 60 seconds (± 10 sec) - Must show: dot placement, units digit identification, carry handoff for at least one example. - Cite the source (PDF + section) in the title slide or closing credit.

Format 2: Printed workbook

Produce a 20-problem workbook with the dot method, designed for a Grade 4 self-learner. Constraints: - 20 problems organised in 4 difficulty levels (5 problems each). - Each level includes ONE fully worked example. - Answer key on the back page. - Title page with cited source.

Format 3: Comparative speed-test data project

Collect baseline times (standard algorithm) and post-training times (dot method) from 5–10 people (family, schoolmates). Constraints: - A table of names (or codes), problem solved, time using standard, time using dot method. - A plot or visual comparison. - A 1-paragraph interpretation of the data — including any surprising findings. - Cite the source PDF.

Why this matters

You're moving from *learner* to *teacher* (or *researcher*). Either role is harder than just doing the problems. The discipline of explaining or measuring forces you to understand more deeply.

What success looks like

A strong project will: - Demonstrate the dot method method accurately (no arithmetic errors). - Be pitched for the intended audience (a 9-year-old for video and workbook; a peer reader for data). - Show the pair's own thinking. - Be clearly communicated.

Working in pairs

- Pair work strongly encouraged.
- Individual allowed for the workbook or video format.
- Groups of 3 acceptable — each member must have a distinct role.

Citation rules

- At minimum **one** citation must appear in your project — visible.
- Format: <PDF filename>, §<section> (e.g. *Vedic Mathematics Batch1.pdf*, §5.4).
- See `sources.md` for the list of available PDFs.

Self-reflection (submitted with project on Day 10)

In 4–6 sentences: 1. What did you set out to learn or make, and what did you actually end up with? 2. What's the most surprising thing you discovered while making this? 3. If you had three more days, what would you do differently?